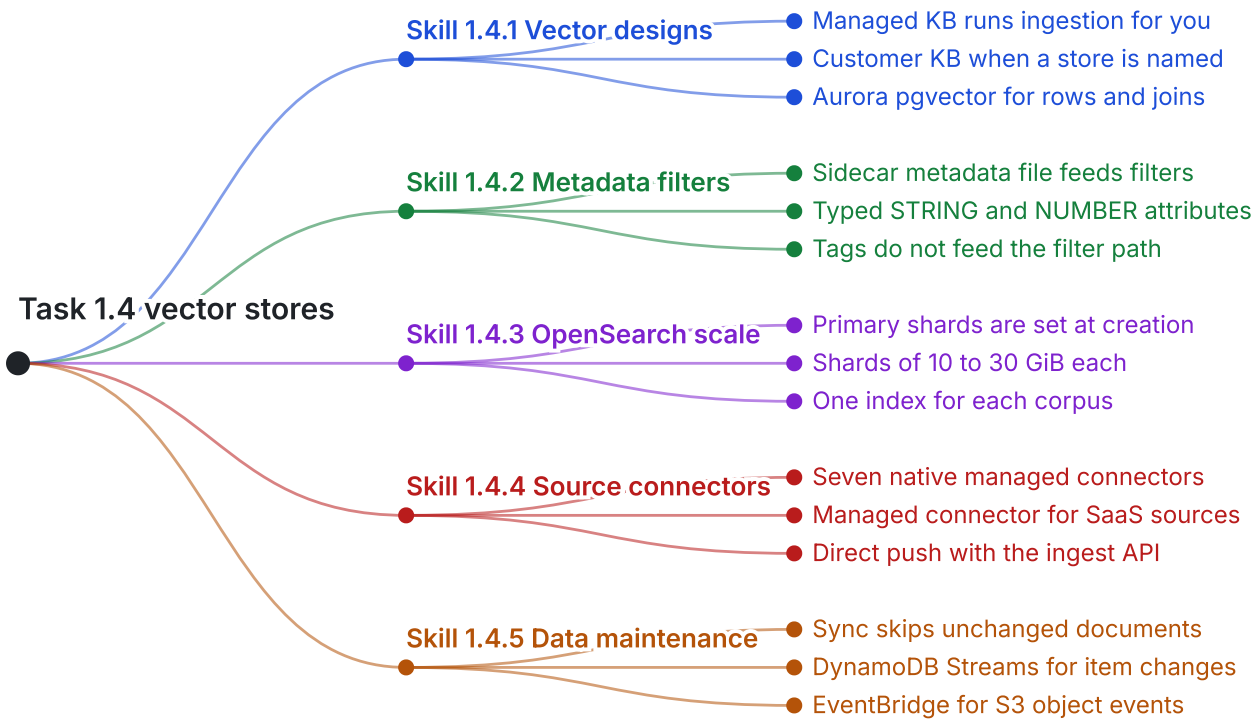
Memory sheet: Task 1.4 Design and implement vector store solutions

Domain 1, weight 31 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
"rows", "join", "transaction", or "same database"	Aurora PostgreSQL with pgvector	The Aurora cluster sits in another account from the knowledge base
"exact codes", "facets", or "full text"	OpenSearch hybrid search	A broad meaning match is enough and no codes or names matter
"rarely queried" with "lowest cost"	S3 Vectors inside a customer-managed knowledge base	Queries are frequent and must return fast, which points to OpenSearch
"private only" or "no public path"	OpenSearch Serverless collection with a VPC endpoint	The store is a managed OpenSearch domain, which needs public access
"within minutes" or "each night"	Event-driven sync for minutes, scheduled job for nights	The stem asks for a full rebuild that re-embeds unchanged content
"filter by author or domain"	Sidecar metadata with typed STRING or NUMBER values	The value lives only in S3 object tags or in the body text
A named store such as Aurora, OpenSearch, S3 Vectors, or Neptune	Customer-managed knowledge base	No store is named and native connectors meet the need, so use managed

Etch lines

1. Find the deciding constraint before you read any option.
2. A managed knowledge base hides the store, so a named store fails that type.
3. Use Aurora with pgvector when rows, joins, or transactions share the vectors.
4. Sync is incremental, so unchanged documents are skipped.
5. Primary shards are set once, when the index is created.
6. A VPC-only managed domain fails for a knowledge base.

Flash questions

- Q: A payment record and its embedding must update in one commit. Which store matches? A: Aurora PostgreSQL with pgvector, in the same account as the knowledge base.
- Q: A design must keep the OpenSearch store private with no public path. Which option works for a knowledge base? A: A serverless collection reached through a VPC endpoint, since a managed domain needs public access.
- Q: A parts finder keeps missing exact product codes in semantic results. Which search mode should it add? A: Hybrid search, with filterable keyword fields in the index.
- Q: A research archive is queried a few times a month, and cost comes first. Which store fits? A: S3 Vectors, inside a customer-managed knowledge base.
- Q: A team saved the attribute file under a name that differs from its source. What happens to that attribute? A: It is ignored, so the filter field never reaches the index.
- Q: A department value is tagged on each file, but retrieval ignores it. Where should the value go? A: Into the sidecar metadata file, since the filter reads only that path.
- Q: A catalog and a support archive use models with different vector sizes. What index design avoids a clash? A: One index per corpus, because each knn_vector field fixes a single dimension.
- Q: Embeddings sit next to items in DynamoDB, but the knowledge base needs a listed store. What is DynamoDB still for? A: It stays the source of truth, and the embeddings sync to a listed store.

Source: The deciding constraint, Skills 1.4.1 to 1.4.5, X versus Y confusable pairs, Read a long question, Distractor patterns, and Drill. AWS docs and the study guide sections above, no practice-test text.